**Mid–Semester Examination**

**Course: PHY101 – Mechanics**

**Maximum Marks: 25     Duration: 1 Hour**

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Answer **all** questions.

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## Section A    ($5 \times 2 = 10$ marks)

*Choose the most appropriate option. Each question carries 2 marks.*

1. A particle is subject to a central force

$$F(r) = -\frac{k}{r^2} + \frac{\alpha}{r^3},$$

   where the constants $k$ and $\alpha$ are positive. For very small $r$, the motion is dominated by:

   (A) The inverse–square attraction

   (B) Sometimes attraction and sometimes repulsion

   (C) A hard–core repulsion from the $\frac{\alpha}{r^3}$ term

   (D) A constant radial acceleration

2. Two smooth beads of masses $m$ and $2m$ are connected by a light spring of natural length $L_0$ and placed on a smooth horizontal table. Initially the spring is stretched to length $L > L_0$ and the system is at rest. At $t = 0$, the beads are released and move toward each other due to the spring force.

   As the beads move, the position of the center of mass of the two–bead system:

   (A) Remains fixed in space

   (B) Moves toward the heavier bead

   (C) Moves toward the lighter bead

   (D) Oscillates about the midpoint of the spring

3. A particle of mass $m$ moves in a straight line with constant velocity $\mathbf{v}$. A fixed point $O$ lies in the same plane of motion but not on the particle's path. At $t = 0$, the particle passes the point on its line of motion that is closest to $O$.

   Which of the following statements about the angular momentum $\mathbf{L}_O$ of the particle with respect to $O$ is correct?

   (A) Both the magnitude and direction of $\mathbf{L}_O$ remain constant in time.

   (B) The angular momentum is zero because there is no rotation.

   (C) The direction of $\mathbf{L}_O$ remains fixed, but its magnitude changes with time.

   (D) Both the magnitude and direction of $\mathbf{L}_O$ change with time.

4. The Sun and Jupiter move under their mutual gravitational attraction. Let $M_S$ and $M_J$ denote their masses, and let the distance between them be $R$. Which of the following statements is correct?

   (A) The Sun remains absolutely stationary, and only Jupiter moves in an orbit around it.

   (B) Both the Sun and Jupiter orbit their common center of mass, and the total linear momentum of the Sun–Jupiter system is conserved.

   (C) Both the Sun and Jupiter revolve around the center of mass, but the total momentum of the system oscillates with time.

   (D) The Sun moves only because it emits radiation, not because of Jupiter's gravitational pull.

5. A person stands inside a lift that is accelerating *upward* with a constant acceleration $a$. At some instant, the person throws a small ball *horizontally* (with respect to the lift) with speed $u$, aiming toward the opposite wall, a horizontal distance $L$ away. Neglect air resistance. Which of the following statements is correct about the ball's trajectory *as seen by the person inside the lift*?

   (A) The ball travels in a straight line and hits the opposite wall at height exactly level with the release point.

   (B) The ball appears to curve *downward* inside the lift and hits the wall below the release point.

   (C) The ball appears to curve *upward* and hits the wall above the release point.

   (D) The ball follows a curved path but strikes the wall at the same height as it was released from.

## Section B (3 × 5 = 15 marks)

6. A particle of mass $m$ moves in a potential

$$V(r) = -\frac{k}{r} + \frac{\alpha}{r^2}.$$

   (a) Show that for certain values of $\alpha$, stable circular orbits cannot exist even for energy $E < 0$.

   (b) Determine the *critical value* of the parameter $\alpha = \alpha_c$ at which the transition occurs between the existence and non-existence of a stable circular orbit. Express $\alpha_c$ in terms of angular momentum $L$ and $m$.

7. A body of mass $m$ is projected vertically upward in a medium in which it experiences a resistance of magnitude $mk^2v^2$. If the initial velocity is $v_0$, show that the body returns to the point of projection with a velocity $v_1$ such that

$$v_1^2 = \frac{g\, v_0^2}{g + k^2 v_0^2}.$$

8. A particle of mass $m$ moves in a plane such that its distance from a fixed point $O$ varies with time as

$$r(t) = r_0(1 + \beta t),$$

where $r_0$ and $\beta$ are positive constants. The particle also moves such that its angular velocity is constant,

$$\dot{\theta} = \omega_0.$$

At $t = 0$, the particle is at $r = r_0$ and $\theta = 0$.

(a) Write down the expressions for the particle's velocity $\mathbf{v}$ and acceleration $\mathbf{a}$ in plane polar coordinates, indicating clearly the $\hat{r}$ and $\hat{\theta}$ components.

(b) Determine the explicit expressions for the radial and transverse (azimuthal) components of $\mathbf{v}$ and $\mathbf{a}$ in terms of $r_0$, $\beta$, $\omega_0$, and $t$.

(c) At what time (if any) does the particle's velocity vector make a $45°$ angle with the radius vector?

(d) At that instant, compute the magnitude and direction (angle with $\hat{r}$) of the acceleration vector. Explain briefly why the acceleration is not parallel to velocity even though $\dot{\theta}$ is constant.

**Total Marks: 25**